v158TH FIGHTER WING
HOME OF THE GREEN MOUNTAIN BOYS
THE GREEN MOUNTAIN BOYS
HQ VERMONT AIR NATIONAL GUARD

STRIKE!

Beyond Visual Range

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Beyond Visual Range (BVR)

Intercepts

Intercepts are a series of maneuvers, whether using a TAC/C2 node, GCI, AWACS, sensor fusion, or dead reckoning, that place the aircraft in a **position** from which a **weapon may be employed**. The type of **intercept geometry** utilized is based on various factors, including but thought not limited to pilot experience and **proficiency**, theatre **context**, available **sensors** or sensor fusions, and electronic warfare **(EW)**.

The intercept problem involves using sensor fusion to **detect** a target, then using geometry to arrive at a position from which the target can be **identified**, and **weapons fired** if necessary. At a minimum, the following tasks should be accomplished during an intercept:

- Positive Identification (PID).
- Positive closure rate.
- Weapons parameters consideration.
- Context consideration.

Baseline Intercept

The **Baseline Intercept** is a foundational technique upon which more advanced tactical gameplans are formulated. The Baseline Intercept consists of gaining an energy advantage over the adversary, as well as building an **optimal intercept course**, often nose-to-nose in theory, but offset in actuality.

Potential energy is a key topic within aviation. Energy, in the context of an aircraft's positioning and velocity, can be derived from a **higher altitude and speed** of the aircraft against Earth's force of gravity and drag. In essence, a pilot should command the aircraft to a higher altitude and airspeed to build an **energy advantage** over the adversary.

Optimal intercept course is a form of lead pursuit that creates a point where the two aircraft are set to collide. Theoretically, given that the pair are co-altitude and co-speed, basic trigonometry can be used to build an optimal intercept course, wherein one aircraft's Aspect Angle (AA) can be added to the other aircraft's AA to add up to 180 degrees.

Time, Distance, & Geometry

Within any BVR engagement, a certain set of pre-defined terms exist to assist and instruct pilots in terms of context, time, distance, and geometry. Here's a breakdown.

- **COMMIT:** Parameters in which to commit to an engagement with a group.
- **NLT COMMIT:** The latest point to commit to an engagement.
- **DR:** Decision Range. A range at which a fighter should decide to commit to staying hot "BANZAI" on the adversary or turn cold "SKATE".
- **DOR:** Desired Out Range. A range at which a fighter should aim to exit the fight at to still have a chance at re-engagement.
- **MAR:** Minimum Abort Range. A range at which a fighter must abort the engagement and "run," or risk being hit by an adversary's weapon solution.

End

You have completed this lesson portion covering BVR.